OLIST E-COMMERCE

Seller Performance Analysis

Challenging a Seller Removal Decision with Data

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Dataset: Brazilian E-Commerce Public Dataset by Olist
Tools: PostgreSQL | Python | Pandas | Matplotlib | Seaborn | Wordcloud | NumPy | Deep-translator

1. Executive Summary

The leadership at Olist had decided to remove sellers with average rating of 2 and below out 5 if the seller has completed a minimum of 10 orders citing that their products were bad.

As the Data Analyst, I challenged this decision and recommended a data-driven investigation be conducted to determine the true root cause of the low ratings.

The investigation examined four potential causes:

- Delivery distance - whether long distances from seller to customer drove poor ratings
- Carrier performance - whether the logistics carrier caused delivery failures
- Seller dispatch time - how quickly the seller handed orders to the carrier
- Customers comments – what keywords are mentioned frequently in their comments

The findings were clear and conclusive. **The seller's low ratings are entirely driven by their own dispatch behaviour - not product quality, not distance, and not the carrier.** The recommendation is not removal, but a structured dispatch time improvement plan.

The dataset can be found at: https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

2. Business Context & Hypothesis

Olist operates a Brazilian e-commerce marketplace connecting sellers to customers across the country. Customer satisfaction is measured through post-delivery review scores on a scale of 1 to 5.

The Challenge

Leadership was considering removing a low-rated seller based on consistently poor customer reviews. As the data analyst, I challenged the decision with the following hypothesis:

> ***"Customers do not separate logistics from product quality when leaving reviews. A late or long-distance delivery creates a negative overall experience - and that frustration gets reflected in the rating, regardless of whether the product itself was good or bad."***

The Investigation Questions

- Did this seller consistently dispatch orders on time?
- Were delivery delays caused by the carrier rather than the seller?
- Do orders travelling longer distances receive lower ratings?
- Do customer review comments mention delivery and shipping rather than product quality?

3. Data Architecture & Pipeline

The analysis was built on a professional data pipeline using PostgreSQL with a dimensional model. The pipeline followed a raw to core architecture ensuring data integrity at every stage.

3.1 Pipeline Overview

Layer	Description	Tables
Raw Schema	All 9 CSV files loaded as TEXT	9 tables — no constraints
Core Schema	Proper data types enforced	5 dim tables, 4 fact tables
Views	Cleaned, layered, reusable	6 analytical views

3.2 Dimensional Model

The dataset was modelled as a star schema with fact and dimension tables:

Table Type	Table Name	Primary Key	Description
Dimension	dim_customers	customer_id	Customer details and location
Dimension	dim_sellers	seller_id	Seller details and location
Dimension	dim_products	product_id	Product attributes and category
Dimension	dim_geolocation	None — deduplicated via view	Zip code coordinates
Dimension	dim_product_category_name_translation	product_category_name	English category names
Fact	fact_orders	order_id	Central transaction table
Fact	fact_order_items	order_id + order_item_id	Line items per order
Fact	fact_order_reviews	review_id + order_id	Ratings and comments
Fact	fact_order_payments	order_id + payment_sequential	Payment details

3.3 Views Created

View	Purpose
fact_orders_clean	Delivered orders with valid complete timeline
fact_orders_reviews	Clean orders joined with review scores
fact_orders_reviews_view	One review per order
fact_orders_reviews_view_clean	One review per order — deduplication applied
dim_geolocation_view	One coordinate per zip code — deduplication applied
seller_dispatch_analysis	Seller dispatch and carrier delivery timeline
seller_vs_platform_view	This seller vs platform average comparison
sellers_reviews_view_clean	Low rated review comments for this seller

4. Data Cleaning

Before analysis, the data was cleaned through a series of systematic checks. All cleaning was applied through views - preserving the raw and core tables untouched.

Check	Finding	Action
NULL delivery dates	645 delivered orders had no review	Excluded via INNER JOIN
Impossible dates	188 orders with delivery before purchase	Excluded via fact_orders_clean view
Duplicate reviews	1,054 rows in duplicate review groups	Deduplicated via DISTINCT ON — kept most recent
Duplicate zip codes	1,000,163 raw rows → 19,015 unique zip codes	Averaged coordinates via dim_geolocation_view
Non-delivered orders	Cancelled, shipped, processing orders excluded	Filtered via order_status = delivered

Final clean dataset: 95,636 delivered orders with exactly one review each - ready for analysis.

5. Key Findings

Finding 1 — The Seller Has a Statistically Significant Low Rating

The seller has an average rating of 1.93 across 14 orders, with 10 out of 14 orders receiving a rating of 1. This pattern is consistent enough to warrant investigation but not immediate removal.

Metric	Value
Average Rating	1.93 / 5.00
Total Orders	14
Orders Rated 1 Star	10 (71%)
Orders Rated 4-5 Stars	3 (21%)
Minimum Review Threshold	10 orders met

Finding 2 - Distance Is Not the Cause of Low Ratings

Analysis of delivery distances showed that some of the lowest rated orders travelled as little as 5km, 9km, and 11km, while orders travelling 331km and 417km received ratings of 4 and 5 respectively. Distance has no correlation with this seller's low ratings.

Finding 3 - The Carrier Is Not Responsible for the Delays

The platform average carrier delivery time is 8.85 days. This seller's carrier took 8.00 days — performing at or above the platform average. The carrier cannot be held responsible for the poor customer experience.

Metric	This Seller	Platform Average	Verdict
Carrier Delivery Days	8.00 days	8.85 days	Normal performance

Finding 4 - The Seller's Dispatch Time Is the Root Cause

The platform average seller dispatch time is 2.74 days. For this seller's late orders, the average dispatch time was 21.11 days — 7 times slower than the platform average. All 9 late orders received a rating of 1. When the seller dispatched within 0 to 5 days, customers rated their orders 4 and 5.

Metric	This Seller (Late Orders)	Platform Average	Difference
Dispatch Days	21.11 days	2.74 days	7.7x slower
Carrier Days	8.00 days	8.85 days	Normal

Metric	This Seller (Late Orders)	Platform Average	Difference
Actual Delivery Days	29.44 days	12.06 days	2.4x longer

Finding 5 - Customers Complained About Delivery, Not Product Quality

Analysis of the 11 customer review comments for low rated orders revealed that the dominant themes were delivery related - not product quality related. Key words appearing in comments included:

Word / Theme	Meaning
delivery, received, receiving	Customers focused on delivery experience
time, long, late, deadline	Timing and lateness of delivery
status, missing, havent, still	Orders not yet arrived — customers chasing updates
waiting, far, inconvenience	Frustration with the wait
return	Customers requesting returns due to non-delivery

Notably absent from all comments: any mention of product quality, broken items, wrong items, or packaging issues.

6. Recommendation

The data does not support removing this seller for product quality issues. Every piece of evidence points to one root cause - the seller's own dispatch time.

Recommendation: Do not remove the seller. Instead, implement a dispatch time improvement plan with the following measures:

- Set a maximum dispatch time threshold of 3 days - aligned with the platform average of 2.74 days
- Monitor this seller's dispatch compliance weekly for 90 days
- Provide the seller with their performance data and platform benchmarks
- Issue a formal warning if dispatch time exceeds 5 days on any order
- Review removal only if dispatch performance does not improve within 90 days

This approach is fair to the seller, protects Olist's revenue, and addresses the actual root cause of customer dissatisfaction.

7. Tools & Technologies

Tool	Purpose
PostgreSQL	Database — raw and core schemas, dimensional model, views
pgAdmin	SQL editor and database management
Python 3.13	Visualisation and text analysis
pandas	Loading SQL results into DataFrames
matplotlib & seaborn	Charts — clustered bar, scatter plot
wordcloud	Word cloud of customer review comments
deep-translator	Translating Portuguese comments to English
VS Code + Jupyter	Development environment and notebook

8. Limitations

- The seller only has 14 orders - a larger order volume would strengthen the statistical significance of the findings.
- Geolocation coordinates are based on zip code prefixes - approximate, not exact addresses.
- Review comments are in Portuguese and were machine translated - minor translation inaccuracies may exist.
- The analysis covers one specific seller - findings may not generalise to all low-rated sellers on the platform.

End of Report